Industrial re-use of a Generic ETCS Hazard Analysis

Dr. Andrew Hussey, Joshua Cobb, Martin Hughes, Yacine Kharoufi, Hubert Grondin & Padma Kodey
Hitachi Rail

Andrew.Hussey@hitachirail.com; Joshua.Cobb@hitachirail.com;
Martin.Hughes@hitachirail.com; Yacine.Kharoufi@hitachirail.com;
Hubert.Grondin@hitachirail.com; Padma.Kodey@hitachirail.com

Abstract

One of the key contributions of the Common Safety Method, also reflected in the EN50126:2017, is the use of Codes of Practice to identify hazards and their associated controls. The UNISIG standards for development of interoperable ETCS systems also incorporate hazard analysis and a hazard log, however this is mainly from the perspective of the components of an ETCS system and their contribution to the ETCS core and auxiliary hazards. Instead, at the overall ETCS System level, a functional analysis is desirable, also reflecting the balance achieved between system and human contributions to failure, so as to properly manage both the EN50126 process as well as demonstrate safety in accordance with SFAIRP. Since this functional analysis is missing from the UNISIG, it needs to be provided for each project where an ETCS system is delivered. In Hitachi Rail, previous projects have created functional analyses as needed to reflect the specific requirements applicable for each customer. However increasingly, the need for standardisation of the analysis, as well as re-use of previous best practice, has become apparent. To accommodate this, a Generic ETCS Hazard Analysis has been developed, which can also be considered to be an emerging Code of Practice for identification of hazards and their associated controls for ETCS systems. This paper is concerned with describing the approach that was taken to developing this analysis, as well as the benefits that this brings, and the remaining challenges to fully realise the complete potential. We describe two case studies where the Generic ETCS Hazard Analysis was used, the optimisations to our process that resulted, as well as the limitations that we observed. The first case study is a project hazard analysis, while the second case study is a training course, incorporating a SFAIRP summary for each hazard in the Generic ETCS Hazard Analysis.

Keywords: ETCS, Code of Practice, EN50126, UNISIG, Hazard Analysis

1 Introduction

The CENELEC standard EN50126 has been commonly applied to railway signalling Systems Engineering as part of the overall CENELEC lifecycle, in conjunction with EN50128 and EN50129. One objective of the EN50126 standard is to provide a framework for the development of safe Signalling Systems, based on the required Safety Reliability of the Safety Functions implemented by that System. Based on the application of the EN50126 process, a functional hazard analysis shall be completed for each project, that enables the identification of hazards and associated controls. Implicitly, the application of EN50126 should lead to more robust systems with fewer faults.

The Rail Safety National Law in Australia imposes the duty of Safe SFAIRP, which is an overarching duty of care to ensure that reasonable measures have been implemented in addressing all potential safety risks. The Safe SFAIRP duty has been interpreted in the ONRSR guideline ONRSR (2021) in respect of how reasonableness shall be judged. One of the commonly accepted interpretations is that where the cost of implementing a safety-related improvement is sufficiently low so as to be less than the benefit x the gross disproportion factor, then the improvement should be implemented. This is irrespective of whether the improvement is needed to achieve the overall THR for the hazard from a purely quantitative perspective.

In this paper, we examine the development of a Generic ETCS Hazard Analysis, as well as its application to a real-world project, to consider how the objectives of both EN50126 and SFAIRP can be more efficiently achieved. We also examine how a Generic ETCS Hazard Analysis can be used to support development of an ETCS training course.

The key benefits of this paper are:

1. sharing experience from development and re-use of a generic hazard analysis in an industrial case study, to accelerate the hazard analysis process and reduce errors;
2. improved understanding of applying Safe SFAIRP in practice;
3. advancement of knowledge of the CENELEC approach, including better understanding of Code of Practice as a Risk Acceptance approach.

2 Acronyms, Abbreviations, and Definitions

2.1 Acronyms and Abbreviations

Acronym	Description
ALARP	As Low As Reasonably Practicable
ATO	Automatic Train Operation
ATSB	Australian Transport Safety Bureau
CoP	Code Of Practice
DMI	Driver Machine Interface
ETCS	European Train Control System
FHA	Functional Hazard Analysis

Acronym	Description
FMEA	Failure Modes and Effects Analysis
FRCMS	Future Railway Mobile Communications System
FTA	Fault Tree Analysis
GAME	Globalement Au Moins Equivalent
GBMS	Global Business Management System
IHA	Interface Hazard Analysis
MEM	Minimum Endogenous Mortality
NCO	Network Control Officer
ONRSR	Office of the National Rail Safety Regulator
PHA	Preliminary Hazard Analysis
RAMS	Reliability Availability Maintainability & Safety
RBC	Radio Base Centre
RSNL	Rail Safety National Law
RTIO	Rio Tinto Iron Ore
SFAIRP	So Far As Is Reasonably Practicable
SIL	Safety Integrity Level
SOM	Start of Mission
SSA	System Safety Assurance
TFFR	Tolerable Functional Failure Rate
THR	Tolerable Hazard Rate
TIRF	Tolerable Individual Risk of Fatality
TMS	Traffic Management System
TVCS	Tunnel Ventilation Control System
VHMI	Vital Human Machine Interface
WSP	Wayside Standard Platform

Table 1: Acronym Table

2.2 Definitions

Following are the definitions useful for this paper as per EN 50126:

2.2.1 Hazard

Condition that could lead to an accident.

2.2.2 Risk Assessment

Overall process comprising a risk analysis and a risk evaluation.

2.2.3 System

Set of interrelated elements considered in a defined context as a whole and separated from their Environment.

3 Literature Survey

ETCS remains Europe's most significant initiative for harmonising railway signalling and ensuring safe, interoperable train operations. As ETCS deployment expands across mainline, regional, and automated rail systems, hazard analysis remains a foundational component of the applicable RAMS (Reliability, Availability, Maintainability and Safety) lifecycle. Industry increasingly emphasises generic hazard analyses — reusable, standardised models developed once and applied across operators, infrastructure managers, and integration projects. This literature review examines academic, regulatory, and industrial perspectives on how generic ETCS hazard analyses are used in practice, their methodologies, and their role in ensuring safe system integration.

Work on hazard identification and risk assessment in ETCS draws from well-established industrial safety approaches originating in aerospace, nuclear, chemical and oil and gas sectors. The use of such methods was initially reviewed by di Tommaso *et al* (2003) as ETCS was being introduced as a railway technology in Europe. A review by Calixto (2022) similarly points to the same risk analysis methods — Functional Hazard Analysis (FHA), Failure Modes and Effects Analysis (FMEA), Fault Tree Analysis (FTA), Preliminary Hazard Analysis (PHA), Interface Hazard Analysis (IHA) — showing how the expectations of railway RAMS standards such as EN 50126/50128/50129 were suitable to analyse ETCS applications. These systematic methods collectively support a generic hazard catalogue, enabling industrial practitioners to classify faults, deviations, and failure modes in a consistent, repeatable manner across ETCS applications.

Although SUBSET 088 of the UNISIG documents a hazard analysis for ETCS systems, this is mainly from the perspective of the ETCS equipment failures and does not cover the overarching functional view of how the ETCS system safely operates within its environment.

Generic hazard analyses for ETCS have been performed by several major rail bodies and research initiatives to streamline safety certifications but are yet to cover the ETCS system in its entirety, rather focusing upon specific ETCS subsystems. Welte (2015) and Transport for New South Wales (2025) have produced generic analyses for ETCS onboard equipment. Rosberg *et al* (2021) produced an application-specific analysis for the Swedish Transport Administration (Trafikverket). Bloomfield *et al* (2016) produced a cybersecurity-focused analysis for ETCS.

The future application of hazard analysis in real operational and industrial environments has been extensively discussed. Calixto (2022) examines how FHA becomes the foundation of defining safety integrity levels, specifying system requirements, and identifying critical failure modes that may lead to derailment or collision. Key insights include:

- Industries rely on FHA early in projects to classify generic hazards;
- ETCS hazard analysis increasingly uses blended qualitative and quantitative techniques;
- Generic ETCS hazard lists developed in earlier deployments significantly accelerate new ETCS upgrades by reusing validated safety arguments.

Calixto also examines how current industrial implementations illustrate emerging issues, especially with ETCS Level 3, where hazards related to train integrity monitoring and Future Railway Mobile Communications System (FRMCS)-based communication are central. These introduce new generic hazard classes related to wireless communication reliability and autonomous integrity assessment that need to be addressed.

Similarly, the use of ETCS as a foundation for Automatic Train Operation (ATO) adds an additional complexity layer to generic hazard models. Tang *et al* (2023) presents a safety analysis integrating FMEA with dynamic modelling for ATO-over-ETCS systems, demonstrating how:

- Generic ETCS hazards extend into the automatic driving domain;
- ATO introduces new system-level hazards (e.g., automation logic failures), which must integrate with ETCS safety constraints.

This supports a growing trend: generic hazard analyses are becoming modular, reused not only across ETCS deployments but also across integrated automation platforms.

Industry reports emphasise the practical value of generic hazard models during ETCS rollouts. RailTech Solutions Ltd. (2024) identifies typical deployment challenges — legacy infrastructure, cross-border variability, and the need for phased introduction — that greatly benefit from generic hazard analysis frameworks. A generic hazard model will allow for:

- Faster conformity evidence for safety assessors;
- Reduction of duplicated hazard analysis across infrastructure projects;
- Consistency in risk acceptance arguments across European corridors.

This aligns with modern ERTMS/ETCS engineering, where uniform hazard models reduce integration risk and enable scalable, repeatable deployments across operators.

Modern ETCS system documentation outlines variations in signalling levels (0, NTC, 1, 2), RBC architectures, and onboard DMI supervision features that influence hazard analysis inputs and parameter sets. These system-level descriptions provide the architectural basis on which industrial practitioners build and reuse hazard analysis artefacts.

Across academic research, engineering practice, and industrial deployment experience, generic hazard analysis has become a cornerstone of modern railway system safety engineering. Its industrial use offers:

- Reusability of validated hazard catalogues and safety arguments;
- Consistency in risk assessment aligned with RAMS and EN 5012x standards;
- Efficiency in ETCS deployment and certification;
- Adaptability across legacy systems, new technologies such as FRMCS, and emerging applications such as ATO.

The literature consistently demonstrates that generic hazard analysis not only accelerates ETCS implementation but also enhances the robustness and harmonisation of the future rail safety landscape.

4 Method for creation of a Generic ETCS Hazard Analysis

The Generic ETCS Hazard Analysis was created via an iterative sequence of workshops with SMEs, to examine use scenarios and extract functions, hazards and associated controls, with particular consideration of key reference projects within Hitachi Rail in Australia. These include a system developed for a major mining freight operator and a system developed for an urban metropolitan railway. The resulting analysis was then subjected to a series of verification checks to assure the correctness and quality of the analysis. Finally, to test the analysis further, various case studies were commenced, using the analysis on a real-world industrial hazard analysis for a project, developing a training course on ETCS and to create a SFAIRP summary for the hazards identified in the Generic ETCS Hazard Analysis.

5 Safe SFAIRP

The obligation to apply Safe SFAIRP in Australia is mandated by the Rail Safety National Law RSNL (2012), but also appears in safety standards such as AS7472 RISSB (2018).

Tim Procter (2019) has considered the interaction between Safe SFAIRP and traditional Safety Assurance activities. He states: "The Safe SFAIRP concept is explicitly included in Australian rail System Safety Assurance (SSA) approaches, but often in a manner that indicates a belief that the implementation of SSA discharges all SFAIRP duties. However, although there is considerable overlap, SSA approaches do not align precisely with Safe SFAIRP principles – particularly when attempting to demonstrate diligence in safety-related decisions-making."

Procter notes that, in practice, the Courts determine what is Safe SFAIRP post-event on a case-by-case basis, with the benefit of hindsight.

Procter concludes that to demonstrate Safe SFAIRP, in addition to the requirements imposed by traditional SSA, it is also necessary to show:

1. *That there is a formal argument as to why all credible, critical hazards have been identified.*
2. *That for each significant hazard all recognised good practice controls are in place, and if not, have been tested for reasonableness, and in the circumstances demonstrated as being unreasonable.*
3. *That further possible practicable controls are considered (even if the risk is considered to be reduced to a 'tolerable' level), and that when considering further precautions, the hierarchy of controls is applied*
4. *That a quality assurance system is in place to ensure all reasonably practicable controls are implemented and remain effective.*

Procter concludes that system safety approaches can end up focusing on maintaining detailed records and following standard activities at the expense of considering what actually constitutes good safety decisions in the specific project context. Rather, as Procter states, to achieve a successful synthesis of SSA and Safe SFAIRP it is critical to remember that SSA is a tool used to achieve a goal, not a goal in itself. The goal is safety, and in the Australian/New Zealand context that is eliminating or (failing that) reducing risk Safe SFAIRP.

Similarly, Steve Dawkins (2021) also notes that *"It is not a Safe SFAIRP argument to list all the treatments identified and imply they amount to Safe SFAIRP, there should also be an argument why these treatments, taken together, represent Safe SFAIRP explicitly".*

Our survey of the literature confirms the need for a Generic hazard analysis and associated Safe SFAIRP argument that provides a basis for re-use and consistency of safety argument across multiple projects.

Under section 46 of the RSNL (2012), Duty Holders are required:

a. to eliminate risks to safety so far as is reasonably practicable; and
b. if it is not reasonably practicable to eliminate risks to Safety, to minimise those risks so far as is reasonably practicable.

The above duties are referred to in the RSNL as the duties to 'ensure Safe SFAIRP'. The persons identified by the RSNL as Duty Holders have a duty of care to ensure Safe SFAIRP.

The concept of Safe SFAIRP is to achieve the best possible Safety outcomes, to the extent that is 'Reasonably Practicable'.

In this context, and under the RSNL (s47), reasonably practicable means that which is, or was at a particular time, reasonably able to be done to ensure safety, taking into account and weighing up all relevant matters including:

a. the likelihood of the hazard or the risk concerned occurring; and
b. the degree of harm that might result from the hazard or the risk; and
c. what the person concerned knows, or ought reasonably to know, about the hazard or risk, and ways of eliminating or minimising the risk; and
d. the availability and suitability of ways to eliminate or minimise the risk; and
e. after assessing the extent of the risk and the available ways of eliminating or minimising the risk, the cost associated with available ways of eliminating or minimising the risk, including whether the cost is grossly disproportionate to the risk.

The ONRSR has published a guideline with respect to understanding the SFAIRP principle ONRSR (2021), the remainder of this Section summarises key observations from that guideline, which is also closely aligned with the Common Law relating to Duty of Care.

What is reasonably practicable is determined objectively. This means that a Duty Holder must meet the standard of behaviour expected of a reasonable person in the duty holder's position and who is required to comply with the same duty.

There are two elements to what is reasonably practicable. A Duty Holder must first consider what can be done – that is, what safety function is possible in the circumstances for reducing residual safety risk. The Duty Holder must then consider whether it is reasonable, in the circumstances to implement those safety functions.

Mitigations shall be selected for a particular Safety Risk (i.e. Hazard Cause) based on:

- the assessed severity of the risk, in terms of likelihood and consequence;
- the reasonably known ways to mitigate that risk;
- the availability/suitability of the known options for reducing risk.

The question of what is reasonably practicable is to be determined quantitatively, and not by reference to the Duty Holder's capacity to pay or other particular circumstances. A Duty Holder cannot expose people to a lower level of protection simply because it is in a lesser financial position than another Duty Holder.

If a particular Duty Holder cannot afford to implement a reasonably practicable risk control, the Duty Holder should not engage in the activity that gives rise to that hazard or risk.

6 Generic ETCS Hazard Analysis Overview

The Generic ETCS Hazard Analysis is a Functional Failure Analysis (FFA) based on the function list that appears in the UNISIG SUBSET 026, in particular Chapter 4 Modes and Transitions and Chapter 5 Procedures, as well as particularisations of the ETCS function that have been recognised and documented in the reference projects. The FFA is separated into UNISIG functions, as per SUBSET 026, functions that are bespoke to Hitachi but which typically appear in our ETCS solutions, and functions that relate to the Interlocking (hence not part of the ETCS, but which are sufficiently novel to warrant specific attention in the Generic ETCS Hazard Analysis. Although titled "Generic ETCS Hazard Analysis" the overall scope of the function analysed therefore does go beyond the core ETCS to cover also other non-core function as well as Interlocking function. The list of UNISIG-related functions covered by the Generic ETCS Hazard Analysis is as follows:

ETCS Function
ETCS SOM operational checks
Localisation
Shunting
Transition from L0 to L2
Transition from L2 to L0
Issue FS MA
Issue SR MA
Supervise MA
Track Circuit / Axle Counter failures
Points failures
Train movement after SPAD
Movement Authority Reduction
Transition between RBC
TSR Management
End of mission
Non-leading engine
Isolation
Virtual Balise Cover

The list of Hitachi-specific functions covered by the Generic ETCS Hazard Analysis is as follows:

Hitachi Function
Blocking for track worker protection
Dwell time management
Hold train command from control centre

Non-stopping zone management
Secure train during passenger/crew exchange
Shunting
Train door opening management
Train stop command from control centre
Transition from L2 to L0

The list of Interlocking-related functions covered by the Generic ETCS Hazard Analysis is as follows:

IXL Function
Axle Counter Unconditional Reset (track block only)
Axle Counter Remote Preparatory Reset (sweep)
Axle Counter Remote Conditional Reset (field technician required)
Forced Release Command for Track Section
Emergency Point Movement
Level crossing management

The fundamental principle of the Generic ETCS Hazard Analysis is that for each new project a review would be made of which functions are applicable and the analysis provided would be customised accordingly. There are 28 Generic UNISIG hazards and a further 19 hazards associated with Hitachi and/or Interlocking function. Examples of UNISIG-related hazards considered in the Generic ETCS Hazard Analysis are:

HZ01	Mission started with incorrect train data
HZ02	Train in ETCS L2 commissioned area operating in Unfitted (UN) mode
HZ03	Mission started with OBU fault
HZ04	Mission started without correctly functioning brakes

Examples of Hitachi-specific hazards considered in the Generic ETCS Hazard Analysis are:

HZ37	Train is stopped for an extended period in a not safe area
HZ38	Train moves when passengers or crew are entering or leaving train
HZ39	Train stops in a non-electrified area and cannot move
HZ40	Track block is created in the wrong location

Examples of Interlocking-specific hazards considered in the Generic ETCS Hazard Analysis are:

HZ44	Warnings not issued for a required period prior to and during the train traversing the crossing
HZ45	Reset of axle counter when the track section is occupied
HZ46	Forced release permitted when there is a train or other vehicle in the released section of track
HZ47	Emergency point movement permitted when there is a train or other vehicle on the points track section

In general, for each ETCS implementation, a separate analysis of the Interlocking function would be required.

7 Case Studies

The first case study involves a project where Hitachi Rail has the safety critical role of European Train Control System (ETCS) supplier. For reasons of confidentiality, details of the System used by the Iron Ore Freight Rail Operator under analysis cannot be shared. However, from an abstract perspective, the high-level system has the following properties:

- Automatic Train Protection (ATP) to manage train movements using fixed signalling block method or virtual moving signalling block system;
- Centralised Train Control Centre, that provides both Interlocking and Vital Authority management functions (Movement Authority, Shunt Authority, Work authority etc) for train separation function;
- Traffic Management System to control train movements in the rail network using inputs such as train timetables and graphs;
- Wayside segment provides continuous communication between the train and the Radio Block Centre (RBC). This is essential for the train to receive real-time updates on track conditions, signal status, and movement authorities.

Generic ETCS Hazard Analysis has been used to prepare the Preliminary Hazard Analysis for the case study and below is the process that has been followed:

- Functional gap analysis was performed between Generic ETCS Hazard Analysis and operator's Concept of Operations to assess functional coverage in the Generic ETCS Hazard Analysis for the operator;
- Generic ETCS Hazard Analysis has been reused for the common functions in the ETCS L2 UNISIG and operator's Concept of Operations. Common functions include SOM operational checks, Transition between ETCS levels, Issue and Supervise MA, Train Movements with TC/points/ AXL Failures, End of Mission, Track Work Authorities etc.
- Based on the Functional gap analysis, additional functions that are applicable to Operator's operations are further analysed in the Functional Failure Analysis (FFA). Functions like Enhanced Shunting – full/partial supervision of train movements in SH mode, Loadout and Dumper functions train supervision in Loadout and Dumper areas, Location Determination System for Localisation function, etc.

- Where applicable mitigations derived from the Generic ETCS Hazard Analysis has been reused and reworded, if required to align with the Operator's requirements.

The second case study involves the development of a training presentation on the re-use of the Generic ETCS Hazard Analysis. Focusing on common risks an operator must consider and what mitigations are to be factored during SFAIRP considerations:

- Each risk in the presentation is identified as a "Failure Mode" in the Generic ETCS Hazard Analysis and is discussed in the training exercise in relation to the associated Hazards, Sub-Function, and Function;
- Furthermore, each risk is dissected into three slides as listed below:
  1. System and procedural controls are identified in the Generic ETCS Hazard Analysis and explained in the presentation on how the risk is mitigated and linked to CoPs;
  2. ETCS Reference Systems are used to prove that the risk has been sufficiently mitigated in practice. The presentation aims to use one metropolitan and one freight reference system per risk, as each service is equipped with their own considerations;
  3. Finally, the residual risk is discussed, along with any additional controls and factors an operator must consider when justifying that SFAIRP has been reached.

8 Outcomes

The objective of using the Generic ETCS Hazard Analysis to support the PHA activity, was to speed up the activity and make the outcome more accurate, accordingly, a key KPI of success is how much of the resulting PHA was based on the Generic ETCS Hazard Analysis. The results indicate that a significant majority of the resulting PHA was based on reused material, hence the use of the Generic ETCS Hazard Analysis can be considered a success:

- 70.5% of the functions in the Generic ETCS Hazard Analysis has been used for the Preliminary Hazard Analysis for the first case study and 29.5% has been added as specific to the Case Study;
- 87% of the Controls (technical/operational) from the Preliminary Hazard Analysis for the first case study is coming from the Generic ETCS Hazard Analysis and 13% of Controls from the Preliminary Hazard Analysis for the first case study is specific to Operator.

The Generic ETCS Hazard Analysis has also been further improved based on the updates made to the Rail Operator's ETCS L2 PHA, for example, shunt related procedural and engineering mitigations have been added. This is as expected, the Generic ETCS Hazard Analysis is a "living document" that shall continue to evolve as new analysis is prepared for each project.

The preparation of the Generic ETCS Hazard Analysis provided significant developmental benefits for junior engineers. Participation exposed them to the breadth of ETCS functions, system requirements, enabling technologies, and associated operational procedures, fostering a system-level understanding of the ETCS ecosystem. This exposure accelerated their technical learning while reinforcing the relationships between functional behaviour, hazards, and risk controls. The activity also served as a structured exercise in safety risk assessment, strengthening their competence in hazard identification, evaluation of mitigations, and formulation of SFAIRP arguments within a real-world railway safety context.

8.1 Challenges Observed

While performing the Operator's FFA, it is observed that certain generic functions of the Generic ETCS Hazard Analysis have not been used by the operator:

- Enhanced Shunting – modifies the standard ETCS shunt to provide boundary supervision;
- Balise related functions – Virtual Balise and Satellite based LDS (Location Determination System). LDS is a SIL 4 system that is installed along the railway network which monitors the locomotive position that is received from the satellite and provided to ATP (Automatic Train Protection). The position is calibrated against fixed reference stations to ensure accuracy. The Virtual Balise acts as a digital trigger that activates when a train's satellite-based LDS matches its real-time coordinates to a specific point on an onboard digital database containing the balise data;
- Because the Operator is a freight operator, Possession Area related functions are used for operations such as for the Load out, which is loading the iron ore from the Mine loading area to the wagons connected to the locomotives and the Dumper, which is dumping (unloading from wagons) the iron ore in the Port area using the Rotary Car Dumper (RCD);
- Some of the modes (Reverse, Passive Shunting, Sleeping, Limited Supervision…) are not yet analysed in the Generic ETCS Hazard Analysis.

Generic ETCS Hazard Analysis Training must be delivered in a concise manner, and as such, the Generic ETCS Hazard Analysis was reviewed and simplified where possible:

- Treatment of ETCS hazards may be homogeneous, whereby a single risk can be defined to create a more concise presentation. Allowing SFAIRP considerations to be written per risk instead of the individual ETCS hazards;
- Most risks cannot achieve SFAIRP purely following ETCS mitigations. To ensure the training exercise still holds value, the training pivoted to what factors an operator must consider as a baseline before justifying SFAIRP.

8.2 Approach to reusability

The Generic ETCS Hazard Analysis is intended as a future Code of Practice for producing ETCS Hazard Analysis and SFAIRP arguments in Hitachi Rail. EN50126 imposes constraints on what is required of a framework or

procedure so it may be considered a Code of Practice, in particular, it must:

- be a set of rules widely recognised in the railway domain. If this is not the case, the CoP shall be justified; and
- be relevant for the control of the hazard in the system under consideration.

Hence to meet the first requirement for recognition as a Code of Practice, the Generic ETCS Hazard Analysis requires additional validation and evidence of its validity, in an industrial setting. To meet the second requirement, the Generic ETCS Hazard Analysis requires clearer "hooks" to enable simpler re-use, more explicitly defining which parts of the analysis can be directly re-used, and which parts shall be customised project by project.

9 Future Ideas

The challenges encountered have led us to consider several future activities to further enhance the usefulness of the Generic ETCS Hazard Analysis:

- Further expansion of the functions that are covered, especially for bespoke Hitachi function;
- Addition of SIL allocation and apportionment to a generic architecture;
- Analysis of interface hazards using a generic architecture and interface model;
- Improvements to re-usability, in particular to add SFAIRP arguments and a risk assessment framework;
- Additional industrial validation.

10 Conclusion

This paper has considered how a Generic ETCS Hazard Analysis can be developed and re-used to speed up the production of ETCS Hazard Analysis for projects, as well as to enable the development of training materials for teaching ETCS RAMS principles.

The key contributions and additions to current learning discussed in this paper were:

1. sharing experience from development and re-use of a generic hazard analysis in an industrial case study, to accelerate the hazard analysis process and reduce errors;
2. improved understanding of applying Safe SFAIRP in practice, through use of the Generic ETCS Hazard Analysis to develop a SFAIRP framework for the applicable hazards;
3. advancement of knowledge of the CENELEC approach by explicitly developing and using a Code of Practice for developing ETCS Hazard Analysis and SFAIRP arguments as well as associated training materials.

11 References

Steve Dawkins (2021): How to Argue Risk is Reduced SFAIRP (https://www.acmena.com.au/insight/how-to-argue-risk-is-reduced-sfairp/), 2021.

ONRSR (2021): ONRSR Guideline Meaning of duty to ensure safety so far as is reasonably practicable, 17 May 2021.

Tim Procter (2019): Compliance vs. Due Diligence: SFAIRP and its Interaction with System Safety and Assurance Approaches, 2019.

RISSB (2018): Railway Operations Management of Change, AS7472, 26 Nov 2018.

RISSB (2020): Complex System Integration in Railways, AS7473, 30 June 2020.

RSNL (2012): Rail Safety National Law (as applicable to each Australian State), 2012.

Tracy A. White (2011): The Language of System Safety Engineering: Loose Language Surrounding ALARP, ASSC 2011.

Dr. Eduardo Calixto (2022): Functional Hazard Analysis (FHA) as basis for safety requirements definition: The ETCS case study, European Federation of National Maintenance Societies, May 2022.

Pasquale di Tommaso, Rosaria Esposito, Pietro Marmo, Antonio Orazzo (2003): Hazard Analysis of Complex Distributed Railway Systems, IEEE Symposium on Reliable Distributed Systems, November 2003.

Jan Welte and Merlin Pokam (2015): openETCS Hazard and Risk Analysis and Safety Case Methodology, ITEA2 Project (Information Technology for European Advancement), February 2015.

Transport for New South Wales (2025): ETCS Onboard Equipment (TS 05266:1.0), 24th April 2025

Tomas Rosberg, Thiago Cavalcanti, Birgitta Thorslund, Erik Prytz and Peter Moertl (2021): Driveability analysis of the european rail transport management system (ERTMS) - A systematic literature review, Journal of Rail Transport Planning & Management, June 2021.

Robin Bloomfield, Marcus Bendele, Peter Bishop, Robert Stroud and Simon Tonks (2016): The risk assessment of ERTMS-based railway systems from a cyber security perspective: methodology and lessons learned, International Conference on Reliability, Safety and Security of Railway Systems (RSSRail), June 28-30 2016.

Ziyue Tang, Ning Zhao, Khalid Kahn and Katinka Wolter (2023): Safety Analysis of Automatic Train Operation Based on ETCS, International Conference on Reliability, Safety and Security of Railway Systems (RSSRail), September 27, 2023.

RailTech Solutions Ltd. (2024): ETCS and the Future of Rail: Building Safer and Smarter Networks, October 31, 2024.